

SIH26168

AI/ML-Based Dead Reckoning for GNSS-Denied Smartphone Positioning

Smart India Hackathon 2026 — ISRO-Sponsored Problem Statement

Project: gudumap | Project Documentation & Technical Summary

1. Problem Framing

gudumap continuously tells the user WHERE THEY ARE, even when GPS is unavailable — inside tunnels, underground parking, or urban canyons — using only phone-grade sensors (accelerometer, gyroscope, magnetometer) with no dedicated external hardware.

Important scope note

This is a positioning system, not a navigation system. The app reports the user's live location and a confidence radius. It does not provide turn-by-turn routing or directions.

2. System Architecture

gudumap is a native Android application (Kotlin, Jetpack Compose) that runs fully on-device with zero backend or connectivity dependency — offline operation is the core design requirement, not an afterthought.

2.1 Processing Pipeline

- Sensor input: accelerometer, gyroscope, magnetometer, and GPS (when available).
- Core estimator: a 6-state Extended Kalman Filter (EKF) tracking position and velocity, combined with Zero-Velocity Update (ZUPT) to detect stillness and a Non-Holonomic Constraint (NHC) that encodes vehicle-motion assumptions.
- While GPS is available: the EKF is corrected directly from GNSS.
- During a GNSS blackout: an on-device GRU machine-learning model predicts displacement for each time window.
- Every ML-predicted correction passes through a kinematic gate that checks physical plausibility before it is accepted into the fused estimate — preventing runaway or non-physical corrections.
- The fused position estimate carries a growing confidence radius the longer the blackout persists.
- An offline map-matching layer (OpenStreetMap data, Coimbatore-only coverage) snaps the estimate to plausible roads/paths before it reaches the live UI.

2.2 What Is Not Part of the Shipped Product

The following were deliberate scope decisions, documented below, and are not part of the shipped application:

- A backend service: an earlier Python/FastAPI prototype exists, but is not used in the shipped app — it was deprecated by design once on-device processing was validated. Only its evaluation-metrics code was reused, purely for offline scoring, not as a runtime dependency.

- An admin/fleet dashboard: this would require a backend aggregating data across multiple users, which is a different product from single-user on-device positioning. It is documented future work, not a gap in the current system.

3. Machine Learning Model

Model	2-layer GRU (Gated Recurrent Unit), exported as gru_io_vnbd.onnx
Runtime	ONNX Runtime, fully on-device inference
Input	20-sample (2-second) window at 10Hz, 6 values per sample (3-axis accelerometer + 3-axis gyroscope, gravity-corrected)
Output	Predicted displacement (dx, dy, dz in metres) for that window
Training data	IO-VNBD — the dataset officially linked on the SIH portal for this problem statement

3.1 Training Data

IO-VNBD consists of 72 real vehicle driving sequences at an average speed of approximately 45 km/h. Smartphone accelerometer and gyroscope readings were logged inside real cars, paired against the vehicle's own CAN-bus / dual-antenna GPS as an independent, more accurate ground-truth reference recorded per session. This is genuinely vehicle-domain data, not simulated or pedestrian data.

3.2 Train / Test Methodology

- Split by whole driving sequence, not by individual window, to prevent correlated-window leakage between train and test: 53 training sequences, 10 validation sequences, 9 held-out test sequences.
- Independently audited against 7 leakage criteria: 6 passed cleanly; 1 was disclosed as a genuine limitation rather than hidden (see Section 4).
- Normalization statistics were independently verified to be re-derivable from the training split alone, confirming no leakage into preprocessing.

3.3 The Pedestrian Model Was Removed

A second model, gru_local.onnx, was trained on OxIOD pedestrian data (phone carried handheld, in-pocket, or in-bag, with lab motion-capture ground truth). Investigation found it was never wired into the live application. It was deleted as dead weight rather than left as unused code. The shipped app is vehicle-only, deliberately, matching the only model that was actually trained, audited, and integrated end-to-end.

4. Accuracy Results and the Heading Caveat

Mean error	11.28 m
Improvement over baseline	54.45% reduction vs. an

	uncorrected baseline
Test set	27,964 real held-out test windows
Verification	Independently spot-checked: real sensor data run through the actual shipped .onnx file produced sane, physically plausible output, confirming genuine (not corrupted or random) weights.

Critical caveat — read before quoting the 11.28 m figure anywhere

The 11.28 m figure was measured using the vehicle's accurate CAN-bus/dual-antenna GPS heading during the blackout window — not the phone's own heading. The shipped app has no CAN-bus access; it relies on the phone's magnetometer-fused heading (Android's TYPE_ROTATION_VECTOR), and the reliability of that heading specifically inside a vehicle or tunnel has not yet been independently measured.

A rough, explicitly labeled back-of-envelope estimate (not a real measurement) suggests an additional 6.6 m to 260 m of lateral error over a 60–120 second blackout, depending on heading quality, against a benchmark median of roughly 250 m of error at 120 seconds with perfect heading.

Correct framing: “11.28 m is the ML displacement-model's accuracy given accurate heading; real-world on-device accuracy also depends on phone heading quality, a disclosed but not yet directly measured factor.”

A real vehicle test with real GPS ground truth is planned but not yet completed — that is the only way to obtain a true end-to-end accuracy number.

5. Key Design Decisions

Each item below was a deliberate, documented scope choice made for stated engineering reasons — not an unaddressed gap or a failure to implement.

- Vehicle-only, not pedestrian: matches the only model that was actually trained and audited end-to-end.
- No backend: connectivity-independence is the core value proposition of the product; a backend would undermine the reason the app exists.
- No admin dashboard: fleet-style aggregation across users is a genuinely different product and is documented as future work.

- Coimbatore-only offline map, not statewide Tamil Nadu: full state coverage was investigated and found to require roughly 3.5GB+ of map data, and would carry real engineering risk — no spatial index in the map-matcher, hardcoded region references across roughly 20 code locations, and no on-demand download capability — for no meaningful benefit to the demo.

6. Development History and Current Status

The system went through substantial iterative debugging. Issues identified and fixed include:

- A hardcoded fallback-location bug — fixed.
- A kinematic-gate feedback-loop bug causing runaway drift — fixed.
- A GPS permission-registration bug — fixed.
- A raw-GPS-speed-spike bug affecting both the displayed speed and the kinematic gate's baseline — fixed.
- General dead-code consolidation across the codebase.

Currently in progress:

- A two-tier UI (a simple view plus an expandable technical detail view) is being implemented.
- One further dead-reckoning drift investigation was in progress as of the last handoff. Its current status should be verified before assuming it has been resolved.

7. Summary

gudumap is a fully on-device, connectivity-independent Android positioning system that fuses an EKF/ZUPT/NHC classical estimator with an on-device GRU model, gated for physical plausibility, to keep tracking a user's location through GNSS blackouts. Its accuracy on the IO-VNBD vehicle dataset (11.28 m mean error, a 54.45% reduction vs. baseline) is real and independently spot-checked, but is conditioned on accurate heading input that the shipped phone-sensor pipeline has not yet had independently measured — a disclosed limitation, with a real-world vehicle validation test planned as next work. Scope choices around the pedestrian model, backend, admin dashboard, and map coverage were deliberate engineering decisions made for documented reasons, not unaddressed gaps.